The Role of Homophily in Response to Labor Market Opportunities: Differences Across Race and Ethnicity

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Abstract

This paper investigates the role that homophily might play in explaining racial/ethnic disparities in the labor market. We find that Black and Hispanic workers are less responsive than White workers to changes in job opportunities, but responsiveness increases when those opportunities present themselves in locations with a higher share own-race population. The analysis makes use of restricted American Community Survey data, accessible through the Federal Statistical Research Data Centers, allowing us to include commuting zones that may otherwise not be identified because of suppressed location information in the public data

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1 Introduction and Background

Long-standing disparities in U.S. labor market outcomes by race are well documented and have been identified as having negative growth consequences for the economy.[1] For example, Bound and Holzer (2000) estimate that different supply adjustments to changes in demand between workers with more or less education or between White and Black workers explains a significant amount of observed differentials in wages across different geographic locations.[2] This paper provides additional evidence of a significant difference in responses across race/ethnicity to labor market opportunities and that considerations of homophily are likely influencing those response differences. The term "Homophily" in this paper refers to the tendency (or preference) of individuals to connect and build social networks based on characteristics they have in common, such as race and ethnicity. This evidence should be interpreted as establishment of a necessary condition for differential response behavior affecting differential labor market outcomes, not a sufficient condition. Identifying factors contributing to differential labor market outcomes helps to point policy makers in the right direction if there's an interest in efficiently addressing these differential outcomes.

A greater ability to chase economic opportunity should improve one's labor market outcomes (for example, see El Badaoui, Strobl, and Walsh 2017; Niebuhr et al. 2009; Davis and Haltiwanger 2014). While not all outcomes from the "Great Black Migration" were positive (e.g.,

[1] For example, see (Antecol and Bedard 2004; Biddle and Hamermesh 2013; Bradbury 2000; Chetty et al. 2019; Engemann and Wall 2010; Fallick and Krolikowski 2018; Zavodny and Zha 2000; Hotchkiss and Moore 2018; Brainard 2017).

[2] Following APA style guidelines, race and ethnic descriptors are capitalized, see https://apastyle.apa.org/style-grammar-guidelines/bias-free-language/racial-ethnic-minorities.

see Black et al. 2015), this phenomenon has been credited with significantly improving economic conditions of Black people from the U.S. South during the early 20th century (Boustan 2015). Burns and Hotchkiss (2020) illustrate greater geographic mismatch between jobs and people among racial minorities than among White workers, offering circumstantial evidence that racial minorities are more constrained in their migration decisions. The implications of constraints for lower economic mobility among minorities is well-documented (for example, see Abramitzky, Boustan, and Matiashvili 2025; Chetty et al. 2020; Durlauf et al. 2024).The goal of this paper is to investigate differences in responsiveness across race to changing labor market opportunities, and whether that responsiveness is sensitive to considerations of homophily, measured by differences in shares of one's own race.

Constraints to migration can take many forms. Wilson (2021) demonstrates that access to information can be an important factor in migration decisions. Cooke (2011) attributes 20 percent of the overall decline in migration rates between 1999 and 2009 to what he calls "secular rootedness," suggesting a cost to migration that might be referred to as a loss of familiarity or networks.[3] Spilimbergo and Ubeda (2004) also establish family ties as a factor affecting migration in their study for differences in migration rates between White and Black people in the U.S. They find that the reason that Black people move less than White people, despite having many factors commonly associated with high migration, is because the Black population, on average, have stronger family ties. Additionally, investigating migration patterns in the 1990s, Frey et al. (2005) confirm that cultural constraints to migration are more prevalent among racial minorities. This is in addition to other differences across race that have long been shown to

[3] Also see Falcettoni and Nygaard (Forthcoming), who find that they need to specify significant utility costs to moving in order to rationalize observed rates of geographic retention.

impact migration decisions, such as access to resources, information, and education (see Greenwood 1975).

An excellent review of the current state of the literature on questions of internal migration is provided by Jia et al. (2023). They cite multiple studies showing that migration is becoming a less important mechanism for demand shock population adjustments. However, our analysis abstracts from these longer trends and asks, holding the dynamic environment constant, whether migration is an even weaker adjustment mechanism for one racial group than another. And, if so, whether homophily plays a role in constraining that mechanism. Our analysis identifies weaker response among racial/ethnic minorities, relative to White, non-Hispanics, to changes in job opportunities across geographic locations. The implication is that worse labor market outcomes among minorities may, at least in part, be the result of greater migration constraints, and that considerations of homophily may be one of those constraints.

2 Methodology

2.1 Empirical Specification

The analysis uses annual data and is performed at the commuting zone (CZ) level.[4] CZs are defined for both rural and urban areas, however dentification of the CZ of a person living in a sparsely populated county is limited for confidentiality reasons in publicly available data. Making use of non-public versions of the American Community Survey (ACS) 1-year estimates through the Federal Statistical Research Data Center, the analysis here is able to provide more comprehensive geographic coverage.[5]

[4] CZ definitions are based on county-to-county commuting patterns; details can be found at https://usa.ipums.org/usa-action/variables/COMZONE#description_section.
[5] Geographic identifiers are suppressed in the 1-year estimates of the public ACS data if the geographic location has a population less than 65,000 people (U.S. Census Bureau 2016).

The empirical model we employ is inspired by that of Amior and Manning (2018b), but differs in that we do not rely on a Bartik instrument (or any other IV approach) to account for the possibility of endogeneity.[6] Instead, we make use of timing, two-way fixed effects (TWFE), and include lagged values of the dependent variable to control for serial correlation.[7] As a robustness test, we estimate a versions with various combinations of fixed-effects. Also, like those who come before us (e.g., Blanchard et al. 1992; Autor et al. 2013; Foschi et al. 2025), we assume a change in population to infer a change in net geographic migration. As pointed out by Foschi et al. (2025), net migration accounts for 60 percent of changes in employment over time suggesting that net migration is more reflective of adjustments to local economic conditions than gross migration.

The main estimating equation is as follows:

$$lnN_g\big|_{e,r,t} = \zeta + \sum_{r=1}^{3}\sum_{e=1}^{3}\left[\beta_{r,e} OwnRaceShare_{g,t}^{r} * RACE_{g,t}^{r} * EDUC_{g,t}^{e} * lnJ_g\big|_{e,r,t-3}\right]$$

$$+ \sum_{k=1}^{3}\left[\theta_k lnN_g\big|_{e,r,t-k}\right] + \omega_1 UR_{g,t-1} + \mu \mathrm{m}_{gt} + \tau_t + g_g + \delta_g + d_g + \alpha_g + \varepsilon_{g,e,r,t}\ , \qquad (1)$$

where,

$lnN_g\big|_{e,r,t}$ = natural log of population in geographic location *g*, of racial group, *r*, with education, *e*, in year *t*;

$lnJ_g\big|_{e,r,t-3}$ = natural log of the number of jobs in geographic location *g*, held by racial group, *r*, with education, *e*, in year *t-3*;

$RACE_{g,t}$ = set of 0,1 regressors indicating White, Black, or Hispanic race/ethnicity;

[6] To use or not to use instruments (even the Bartik) has been a long-running debate. For example, see these key references: Bound et al. 2004; Goldsmith-Pinkham et al. 2020; Borusyak et al. 2025.

[7] Note that recent criticisms of TWFE specifications are in the context of difference-in-difference treatment models (Imai and Kim 2021). That is not the context in which TWFE is used in this paper. These steps to account for potential endogeneity are discussed further in Section 3.5.

$EDUC_{g,t}$ = set of 0,1 regressors indicating high school, some college, or college plus;

$lnN_g\big|_{e,r,t-k}$ = one, two, and three year lags of natural log of population in geographic location g, of racial group, r, with education, e;

$OwnRaceShare_{g,t}^{r}$ = share of the population in CZ g of race r (r = White, NH; Black, NH; Hispanic);

$UR_{g,t-1}$= one-year lagged unemployment rate for CZ g as a measure of disequilibrium (e.g., see Devaraj et al. 2017);

m_{gt} = migrant shift-share;

τ_t = year fixed effect;

g_g = CZ fixed effect;

δ_g = CZ-specific distance (population-weighted centroids) from next nearest CZ;

d_g = CZ population density in 1990 (total population in CZ divided by total land area of CZ);

α_g = CZ-specific amenities; and

$\varepsilon_{g,e,r,t}$ are robust standard errors, clustered by CZ level.

We classify jobs by education and also by race/ethnicity based on Hellerstein, Neumark, and McInerney (2008)'s finding that an absence of the availability of jobs, generally, is not enough to explain lower employment rates of Black workers, but it's the absence of jobs *available to Black workers* that matters -- accounting for the distribution of jobs only by education level would ignore this point.[8] This race/education specific number of jobs is our measure of job opportunities in a specific geographic location. One might also argue that a measure of job vacancies would better reflect job opportunities, but because of the importance of

[8] Also see Wozniak (2011) for a unique study documenting the systemic challenges faced by low-skilled Black men, relative to their equally low-skilled White counterparts.

identifying race-specific job opportunities, it is not possible to use vacancies for this purpose since it is illegal to specify race when advertising a job opening.[9]

Lagged values of the dependent variable ($lnN_{g_{e,r,t-k}}$, k=1,2,3) are included as regressors in order to remove from the error term unobserved factors that are expected to be serially correlated with both population and jobs (Baerg et al. 2018). Note that Nickell (1981) warns of significant bias when including lagged values of the dependent variable in an autoregressive regression with fixed effects. However, his analysis is concerned with bias on the autoregressive terms, not on the estimated coefficients of other regressors in the model. Additionally, Keele and Kelly (2006) show that if the model is dynamic (which this is), including lagged terms of the dependent variable results in less bias than if the lagged terms are omitted.

Further, de Chaisemartin and D'Haultfœuille (2023) identify a primary reason why two-way fixed effects may produce biased results in the presence of heterogeneity in the relationship of interest. This is not as serious a concern here as in other applications since our relationship of interest (between job opportunity and population growth) is allowed to vary by both education and race (and own race share) -- perhaps two of the most important sources of heterogeneity. Also, Imai and Kim (2021) warn that the ability of two-way fixed effects to identify unobserved time and location confounders of the relationship of interest may fail if the confounders are not additively linear. We estimate a fully flexible fixed-effects model to assess whether the estimated two-way fixed effects results are robust.

[9] There is a growing body of research using online vacancy data, such as Glassdoor or Vault (for example, see Kureková, Beblavý, and Thum-Thysen 2015). Additionally, the Bureau of Labor Statistics makes available measures of job openings (vacancies) in their Job Openings and Labor Turnover Survey (JOLTS). But these data are available only by industry or broad Census region, not both. In addition, occupation is more reflective of educational requirements than industry, which will employ workers of a much broader range of educational attainment. But more importantly neither online vacancy data nor JOLTS identifies race-specific job opportunities.

Since we are particularly interested in the role that homophily plays as a constraint on migration, we fully interact the number of education/race-specific jobs in each commuting zone with the share of the commuting zone's population that is of race r ($OwnRaceShare_{g,t}^{r}$). The relationship between education/race specific jobs and education/race specific population will be evaluated at different values of *OwnRaceShare*. If homophily plays a role in responses to job opportunities, we expect a positive influence of *OwnRaceShare* on the relationship between job and population changes. While this result would not be a sufficient condition, it is a necessary condition to draw this conclusion.

The regressors, δ_g, d_g, α_g, and m_{gt}, are labor supply controls. Of course, with both year and location fixed effects included in the model, the non-time varying regressors (population density and amenities) drop out of the equation.

Amior and Manning (2018b) point out that if population immediately (or within three years, in our case) adjusts to job opportunities, then the marginal effects on $\left(lnJ_g\right)_{e,r,t}$ would be equal to one, as there would be no deviation from equilibrium. Observations are weighted by lagged CZ population shares, which are computed using the Census counts of population when available, and Census estimates otherwise.

The migrant share (varying by time and CZ) is included as the presence of a large share of migrants may either attract (as an enclave) or detract (as competition) population growth. Since the migrant share may be endogenous, migrant shares are replaced with a standard Bartik shift-share instrument (for example, see Card 2001); details of construction of the migrant shift-share regressor are included in the appendix.

The analysis is restricted to CZ/race/education observations that have non-zero values for current and lagged values of population and jobs. The reason for this restriction is that we do not

know whether a zero race/education combination in a specific location is a true zero, or whether that location was simply not sampled that year of the survey. We further restrict the sample to create a balanced panel; a specific CZ/race/education combination has to have non-zero observations for each year of the analysis to be included. This means that if one race/education combination is missing for a CZ, but another race/education combination does not have missing values, the CZ is retained, but only for those race/education combinations with complete data through the time period. The unbalanced data is used as a robustness check, yielding results that are not appreciably different; those results can be found in Table A2 of the appendix.

2.2 Data

The analysis in this paper makes use of the restricted American Community Survey (ACS) 1-year estimates from 2005-2019. These data are accessible through the Federal Statistical Research Data Centers, allowing us to include respondents from commuting zones that may otherwise not be identified because of suppressed geographic information in the public data. The ACS is a nationally representative cross-sectional survey and has been administered annually since 2005 to about 2 million households and is well suited for subnational analyses.[10] We make use of the ACS-provided individual weights when creating aggregate variables and the analysis in this paper is confined to the 16-64 year old population, excluding the armed forces. In addition, the analysis excludes less than high school as an education group and individuals whose race is classified as Asian or "other."

For each year, the median education level is determined (using person weights) for each detailed occupation in order to classify each job by its educational "requirement."[11] Table 1

[10] https://www.census.gov/acs/www/methodology/sample-size-and-data-quality/sample-size/
[11] Using the median is preferred since several occupations had multiple "modes." We do not restrict this determination to certain ages and we allow the median to change over time.

reports the distribution of occupations across median education. Most occupations have a median education level of a high school degree or some college. Just over half of a percent of all occupation codes have a median education level of less than a high school degree.[12]

[Table 1 about here]

Table 2 reports the distribution of the 16-64 year-olds in the ACS across race/ethnicity for each educational group. White, non-Hispanics make up the largest share in all education groups, except those with less than a high school degree. The shares of Hispanics decline uniformly in educational attainment whereas the shares of White, non-Hispanics increase uniformly in education; the share of Black, non-Hispanics is lower among college educated than shares in other educational groups.

[Table 2 about here]

The number of job opportunities for a person of a certain race with a certain education level in a certain geographic location is proxied by the number of jobs requiring that education level in that geographic location held by workers (aged 16-64) of that race. The population in each geographic location for each race and education group in each year is simply the count of the number of people (aged 16-64) of race *r*, with education *e*, in location *g*. Table 3 reports the unweighted means of the variables used for the analyses. Of course, with both year and location fixed effects included in the model non-time varying regressors drop out of the estimation. Overall, White, non-Hispanic workers are located in CZs that are less densely populated, have

[12] Hispanic workers are overwhelmingly represented among those with less than a high school degree (see https://www.census.gov/data/tables/2022/demo/educational-attainment/cps-detailed-tables.html). The results in this paper are generalizable only to those in the population who have at least a high school degree and identified as White or Black, non-Hispanic or Hispanic.

higher amenities, and have lower rates of unemployment. Hispanic workers are located in CZs with much higher shares of migrant populations.

[Table 3 about here]

3 Results

3.1 Overall Marginal Effects

Table 4 contains the marginal effects for the race/education specific change in log jobs on the race/education change in log population from an OLS estimation of equation (1). The table compares specifications including contemporaneous vs. three-year lag of log jobs, as well as specifications not controlling for serial correlation with results controlling for serial correlation (i.e., including lags of the dependent variable). Note that replacing contemporaneous with lagged values of log number of jobs reduces the marginal effect only slightly, but controlling for potential serial correlation substantially reduces the point estimates. All specifications in Table 4 are weighted by CZ population shares and include two-way (separate year and CZ) fixed effects.

[Table 4 about here]

The marginal effect is interpreted like an elasticity. For example, a marginal effect estimate of *m* means that a one percent change in race/education-specific jobs in a CZ is associated with an *m* percent change in population in the CZ for that race/education group, three years later. These are overall effects by race and education group, at average *OwnRaceShare* values. Because of disclosure restrictions, parameter coefficients from the estimations are not reported. These marginal effects being less than one is an important point made by Amior and Manning (2018b); there is never perfect adjustment of population to changes in job opportunities and this is why differences in labor market outcomes persist across geographic locations. Analogously, differences in adjustments across race and education is part of the reason for

persistence in labor market outcome differences across race and education groups, even within a geographic location.

Before looking at differences across ethnic/racial groups, it's of interest to note that the marginal effects mostly conform to conventional wisdom about greater migration among more educated individuals who would have, all else equal, greater access to information and resources to facilitate a response to job opportunities in another location (e.g., see Greenwood 1975; Malamud and Wozniak 2012; Wozniak 2010). For example, Black (White), non-Hispanic college graduates are two and a half times (100 percent) more responsive than those with only high school attainment. Hispanic workers with a college degree are nearly 30 percent more responsive than those with a high school degree. Note, however, that Black and Hispanic groups with some college are slightly more responsive than those in the same group with a college degree.

We also see in Table 4 that overall, and within education groups, Blacks and Hispanics are less responsive to changing job opportunities than are Whites. For example, a one percent increase in jobs available to White workers in a CZ is associated with a 0.0095 percent increase in the White population, whereas the analogous response among Black and Hispanic workers is 0.0034 and 0.0057 percent, respectively. These differences in responsiveness can have very real impacts on labor market outcomes. For example, Bound and Holzer (2000) estimate that lower population adjustments among Black and less educated workers to an exogenous demand shock of, say, 10 percent, contributes roughly three percent to the wage differences observed across geographic location between Black and less educated workers and White and more-educated workers, respectively.

3.2 Sensitivity to Own Race Share

Strong social and cultural ties have been found to be an important determinant of an individual's willingness (or ability) to migrate in response to a negative labor market event (Huttunen et al. 2017; Zabek 2019). Kosar, Ransom, and van der Klaauw (2019) find that that strong (and growing) preferences for family and local cultural norms (homophily) partially explain the long-run decline in migration rates in the U.S. A graphical analysis of Facebook connections illustrates how powerful connections from historical events, like the Great Migration in the early 20th century, can dictate geographic connectedness today (Bailey et al. 2018, also see Badger and Bui 2018).

Also, Ananat, Shihe, and Ross (2018) find that as the share of a worker's race in a local area increases, the employment density wage premium for that worker increases, providing yet another reason why we might expect minorities to respond more to employment opportunities in areas with higher own-racial shares (at least in densely population CZs). This section explores the role of just one of many possible factors that might be playing a role in weaker responsiveness of minority workers to changes in job opportunities -- the share of own-race population in the location offering job opportunities.

To investigate the importance of own-race population share, equation (1) includes the interaction of own-race/ethnic population shares with the education and race modifiers on the number of jobs regressor. This allows the race/education specific relationship between $lnJ_{g,e,r,t-3}$ and $lnN_{g,e,r,t}$ to vary across different levels of concentration of racial minority population within the CZ. The hypothesis, that own-race share plays a role in migration decisions, would be supported if the relationship between the percentage change in jobs and population is stronger when the CZ has a higher share of own-race population. This would

suggest that homophily is playing a role in measured responsiveness. Table 5 contains marginal effects for the preferred specification to test this hypothesis (i.e., weighted with CZ population shares, three-year lag of $lnJ_g\big|_{e,r}$, balanced panel, two-way fixed effects, and correction for serial correlation) at different points in the distribution of race shares in the CZ.

[Table 5 about here]

The first thing to notice in Table 5 is that the differences in marginal effects across education groups perform as expected for all points of the own-race share distribution. Additionally, the weaker response of Black and Hispanic workers, relative to White workers, is found across the own-race share distribution.

The additional information contained in Table 5 is that responsiveness of the education/race population to increasing job opportunities in a CZ increases when workers' own-race share is higher. For example, a one percent increase in job opportunities for Black workers in a commuting zone where the Black race share is in the national 75th percentile of Black race shares is associated with an increase in the Black population by 0.0034 percent, whereas the response is only an increase of 0.0008 percent in CZs in the lowest quartile of Black race share (four times lower than at the 75th percentile).

The results in Table 5 also indicate that responsiveness to job opportunities is more sensitive to own-race share among Hispanic workers than among Black workers. And the difference in responsiveness across quartiles among both Hispanic and Black workers is greater than among White workers. This pattern is consistent with Stuart and Taylor (2021) who find that, historically, social/cultural networks are much more important for determining migration patterns of Blacks than for Whites, especially with the expectation of increased job opportunities.

To put the importance of these differences across the own-race share distribution into perspective, note that, among Black workers overall, responsiveness to a job opportunity in a location in the 75th percentile of Black race share is six time higher (on average, across all education levels) than in a location at the 25th percentile -- this is two times greater than the difference between responsiveness among Black college graduates relative to Black high school graduates.

3.3 Validity test - responsiveness to other race shares

If our interpretation of the way in which a group's own-race share is modifying the relationship between changing job opportunities and population within a commuting zone is correct, then we should observe that a group's response to its own race share is also greater than its response to changes in other groups' race shares. So, as a validity test, we estimate a slightly modified equation (1) to confirm this is the case. The new specification is depicted below:

$$lnN_g\big|_{e,r,t} = \zeta + \sum_{r=1}^{3}\sum_{e=1}^{3}\left[\beta_{r,e}^{W} WRaceShare_{g,t} * RACE_{g,t}^{r} * EDUC_{g,t}^{e} * lnJ_g\big|_{e,r,t-3}\right]$$
$$+ \sum_{r=1}^{3}\sum_{e=1}^{3}\left[\beta_{r,e}^{B} BRaceShare_{g,t} * RACE_{g,t}^{r} * EDUC_{g,t}^{e} * lnJ_g\big|_{e,r,t-3}\right]$$
$$+ \sum_{r=1}^{3}\sum_{e=1}^{3}\left[\beta_{r,e}^{H} HRaceShare_{g,t} * RACE_{g,t}^{r} * EDUC_{g,t}^{e} * lnJ_g\big|_{e,r,t-3}\right]$$
$$+ \sum_{k=1}^{3}\left[\theta_k lnN_g\big|_{e,r,t-k}\right] + \omega_1 UR_{g,t-1} + \mu \mathrm{m}_{gt} + \tau_t + g_g + \delta_g + d_g + \alpha_g + \varepsilon_{g,e,r,t}\ , \quad (2)$$

where $\{W,B,H\}RaceShare$ corresponds to the share of the CZ population that is White, Black, and Hispanic, respectively. Basically, the regressor *OwnRaceShare* from equation (1) is replaced by *WRaceShare*, *BRaceShare*, and *HRaceShare* in equation (2). The question is, for example among Black workers, to what degree does *BRaceShare* modify the relationship between jobs and population relative to the impact of *WRaceShare* or *HRaceShare*. If, in this example, *BRaceShare* has the largest impact, this

affirms the greater responsiveness estimated for higher levels of a group's own-race share. Results from this analysis are found in Table 6.

[Table 6 about here]

In order to perform this validity analysis, we needed to eliminate a few CZs that didn't have racial representation across all races; this creates a new unbalanced sample. Column (1) of Table 6 repeats the baseline marginal effects from Table 4 for each race, just as a point of comparison for the new sample. Column (2) of Table 6 reports the marginal effects from the original specification of equation (1) except using the new, unbalanced, sample (with fewer CZs) used for this validity analysis. The marginal effects in columns 1 and 2 are very similar and exhibit the same pattern across race groups. Therefore, differences between the original sample and this new, smaller, unbalanced sample are not a concern for the validity test.

Column (3) is a different marginal effect than those reported in Tables 4 and 5, is also from the original specification of equation (1), but with the new restricted sample. This column contains the marginal effect for the regressor "*OwnRaceShare*" (at average values for education and log number of job values) for each racial group. This marginal effect is reported to illustrate the same pattern of greater responsiveness by Blacks and Hispanics to changes in *OwnRaceShare* in the original specification. In other words, responsiveness of Black and Hispanic workers to job opportunities in a CZ increases with own-race share in that CZ. Now to the actual validity test.

The marginal effects in columns (4)-(6) are analogous to those in column (3) except estimated using the specification from equation (2), which replaces a group's own-race share with measures of all three groups' race shares. Note that these marginal effects are evaluated at average education levels and average CZ log job values. We see that for each group, their own

race share is more important in modifying the relationship between job opportunities and population than either of the other two groups' race shares (noted by the bolded numbers along the diagonal). For example, a one percentage-point increase in the Black race share increases the Black population response by 0.0171 percent, whereas a one percentage-point increase in the White (Hispanic) race shares increase the Black population response by 0.005 (0.006) percent, respectively. Also note that White workers are only 11 percent more responsive to changes in their own race share than to changes in either Black or Hispanic race shares. However, Black workers are roughly 100 percent more responsive to their own race shares and Hispanic workers are between 50 and 70 percent more responsive to changes in their own race shares than to changes in population shares of other races.

These results support the conclusion drawn from the results in Table 5 that each race group is more responsive to changing job opportunities when they occur in CZs in which they have greater representation.

3.4 Robustness

A number of robustness checks were performed. These results are not included in the paper due to disclosure concerns. The baseline specification of equation (1) was estimated using different lag values for number of jobs without any appreciable difference in results. The two-way fixed effects version of the model is preferred to excluding CZ fixed effects since various CZ-specific regressors alone did not seem to be able to adequately control for the persistent nuances across location that might be expected to explain variations in education/race specific population changes.

Additionally, different combinations of the fixed effects were estimated. In contrast to the results presented here for the two-way separate year and commuting zone fixed effects, a

grouped (year-by-commuting zone) fixed effect was estimate with no appreciable difference in results (of course, all other regressors dropped out of the regression). Including a year-only fixed effect produced marginal effects that were nonsensical from an economic perspective. This exercise makes it clear that place-specific fixed-effects are the most important control.

In the Appendix we present marginal effects for unweighted regressions and for an unbalanced version of the sample. The arguments for using weights and focusing on the balanced panel results are presented in the Appendix. The results from those specifications were not materially different than those presented in the text.

3.5 Caveats about Causality

Estimating the relationship between employment growth and population growth is subject to classic endogeneity concerns. While job growth can "pull" people to a location to take advantage of the job opportunities, population growth can, in turn, result in employment growth through increased demand. Likely because of the relatively short period of time of the analysis, and the commonality of factors affecting the "shift" component across disaggregated race and education groups (see Borusyak et al. 2025), a standard shift-share instrument (of varying constructions) is not viable.

As an alternative, we make use of timing, both year and CZ fixed effects, and include lagged values of the dependent variable to control for serial correlation. The timing factor involved lagging the variable of interest by three years. It would be hard to imagine that population growth (the dependent variable) three years in the future, *all else equal*, has an impact on current job growth.

One could argue, however, that unobserved factors across CZs drive both employment and job growth -- this is controlled for through the inclusion of place-based (CZ) fixed effects

(akin to the "share" portion of a shift-share instrument) and other regressors that vary across place and over time (i.e., lagged unemployment rate and migration shift-share).

It is also possible that there exists significant serial correlation in population growth, which would lend support to an argument that population growth in the future (which is highly correlated with current population growth) is influencing current job growth. To control for this possibility, we include as regressors one-, two-, and three-year lags of population growth to remove contemporaneous correlation that may still be present in the error term. The effect of these controls can be seen in the different columns of Table 4. It's clear that the correction for serial correlation is more impactful in eliminating spurious correlation from the error term than lagging job growth.

However, to the extent that there remain factors in the error term affecting both job opportunities and population growth, the results in this paper can only be interpreted as relational, not causal.

4 Conclusions and Policy Considerations

The analysis in this paper presents evidence of different responses by education and race to changing job opportunities across commuting zones (CZs). The relationship between the change in education/race specific job opportunities in a location and the change in education/race specific population is larger among White, non-Hispanic workers than it is for Black, non-Hispanic workers or for Hispanic workers. Additionally, the results suggest that homophily may play a role in constraining ethnic/minority response to changing labor market opportunities -- responsiveness increases as the share of a group's own race share in the population increases.

The stronger response when job opportunities arise in locations with greater minority representation is not entirely unexpected. Others have found that racial and ethnic minorities

experience significant gains from social and cultural networks that are accessible when living in close proximity with one another (e.g., Montgomery 1991; Edin, Fredriksson, and Åslund 2003; Elliott 2005). This would suggest that efforts directed toward decreasing disparate labor market outcomes should focus on adjusting the human capital of minorities (e.g., by improving educational opportunities) to better match the occupational demands of the area, or by improving economic opportunities that better match the educational attainment of the population, rather than necessarily promoting migration.

One might argue that differences in response to job opportunities simply reflect individual differences in the calculation of the benefits and costs of moving. However, attributing a consistent difference in behavior across race or gender to simply differences in preferences ignores the potential for broader systemic factors affecting the calculus of those costs and benefits. For example, Dickerson (2007) finds that employment outcomes are worse for Blacks in segregated cities, suggesting that geographic concentration may be harmful for economic outcomes of minorities, and that easing other migration constraints might prove useful for improving labor market disparities. Additionally, Bove and Elia (2017) find that increased ethnic diversity is associated with higher economic growth, adding to the benefit side of any efforts to incentivize majority to minority migration.

Picard and Zenou (2018) provide a theoretical model showing how minority workers, faced with a mismatch of location and jobs, could benefit from a variety of policy approaches. Place-based policies, such as neighborhood regeneration (which provides incentives for majority workers to move there providing improved networking contacts) and establishment of enterprise zones (attracting firms providing additional employment opportunities) are ways in which specific geographic locales can attract both residents and firms. Contrastingly, people-based

policies, such as the Moving to Opportunity programs, provide housing subsidies in order to improve outcomes by moving people closer to jobs.[13] Incentivizing people to move can be a tall order (for example see Harrison and Raice 2018) and may results in unintended reactions from receiving populations (Derenoncourt 2022). However, Cáceres-Delpiano et al. (2021) and Cáceres-Delpiano et al. (2023) offer evidence of a broader benefit to policies that potentially move people out of their comfort zones; they find that exposure to different regional peculiarities allows individuals to identify cultural and social commonalities that span geography, increasing the willingness to migrate.

The potential conflict in policies focused on *either* people *or* place is long-standing in the urban literature, described in a phrase coined by Winnick (1966) -- 'Place Prosperity vs. People Prosperity' (also see Bolton 1992; Partridge and Rickman 2007). This leads to a potential role for indirect policies, such as improving public transportation or access to information (see Waldrip et al. 2015; Wilson 2021) for improving employment outcomes among minorities. In their extensive review of the migration literature, Jia et al. (2011) also highlight the importance of information for facilitating migration as a mechanism to improve individual labor market outcomes. They call for a better understanding of the role that community and social ties play in those migration decisions. The analysis and results in this paper are a step in that direction.

[13] Also see Mueller 1981, who describes the apparent success of a relocation assistance program in the 1970s in getting people to move to better job opportunities, even those who expressly indicated they didn't want to move.

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Table 1 Distribution of occupations across median education of those employed in the occupation 2005-2019.

Median Education in Occupation	Percent of Occupation codes across years
Less than high school	0.58%
High school degree only	35.02%
Some college	39.24%
College degree and above	25.17%

Notes: Authors calculations using restricted ACS.

Table 2 Distribution of 16-64 year old population across race/ethnicity by educational attainment, 2005-2019.

	Percent of Education Category			
	White, NH	Black, NH	Other, NH	Hispanic
Less than HS	43%	12%	6%	39%
HS degree	66%	14%	4%	16%
Some College	69%	13%	5%	13%
College degree or more	75%	8%	10%	7%

Notes: Authors calculations using restricted ACS. Row totals may not sum to 100 due to rounding.

Table 3 Means and standard deviations for variables used in the analyses.

	Average Census Tract Characteristics								
	Population (N_g) (000)	Employ-ment (J_g) (000)	Own Race Share	Lagged unempl. rate	Miles to the next closest CZ	Share of population that is migrants	Amenity scale	Population density in 1990	Observations
White, NH	47.61	40.28	0.7872	6.963	47.99	0.2438	0.1527	91.13	26000
	[112.6]	[93.71]	[0.1759]	[3.649]	[16.28]	[2.359]	[0.3642]	[243.8]	
High School	38.29	33.76	0.7867	6.965	47.97	0.2418	0.1579	90.82	8600
	[69.55]	[60.2]	[0.177]	[3.674]	[16.28]	[2.359]	[0.382]	[243.4]	
Some College	51.55	48.99	0.7868	6.947	50	0.2472	0.1582	90.81	8600
	[103.6]	[108.5]	[0.1769]	[3.638]	[16.28]	[2.361]	[0.3825]	[243.4]	
College and Above	53.06	38.08	0.7881	6.977	50	0.2424	0.1419	91.77	8500
	[149.8]	[104.1]	[0.1737]	[3.635]	[16.29]	[2.356]	[0.3243]	[244.6]	
Black, NH	16.66	13.62	0.1429	7.909	46.56	0.2162	0.0306	160.8	13000
	[40.13]	[33.8]	[0.136]	[3.111]	[14.61]	[2.155]	[0.0362]	[328.1]	
High School	15.49	14.57	0.1347	7.87	46.37	0.1615	0.0327	151.3	4700
	[34.8]	[30.68]	[0.134]	[3.121]	[14.44]	[2.116]	[0.0414]	[315.7]	
Some College	20.76	16.47	0.1411	7.886	46.46	0.2153	0.0311	158.3	4500
	[47.49]	[42.5]	[0.1354]	[3.11]	[14.48]	[2.159]	[0.0356]	[300]	
College and Above	13.32	9.097	0.1551	7.985	46.93	0.2852	0.0276	175.4	3800
	[36.33]	[24.18]	[0.1386]	[3.099]	[14.97]	[2.196]	[0.0291]	[347.1]	
Hispanic	15.56	17.4	0.1342	7.181	50.25	0.5986	0.091	136.1	15000
	[64.55]	[79.57]	[0.1666]	[2.974]	[16.8]	[2.508]	[0.2291]	[300]	
High School	15.33	22.38	0.1246	7.123	49.49	0.4814	0.1086	118.1	6100
	[66.45]	[100.3]	[0.159]	[3.085]	[16.6]	[2.497]	[0.2678]	[283.6]	
Some College	18.04	17.49	0.1356	7.127	50.67	0.6023	0.0898	133.1	5100
	[74.34]	[75.45]	[0.1666]	[2.937]	[16.82]	[2.504	[0.2155]	[300]	
College and Above	12.56	9.257	0.1477	7.349	50.9	0.7824	0.0641	200	3800
	[43.69]	[32.89]	[0.1774]	[2.833]	[17.05]	[2.521]	[0.169]	[349.7]	

Note: American Community Survey 2007-2019, balanced panel, unweighted. Numbers of observations are rounded for reasons of confidentiality.

Table 4 Marginal effects of a change in the log education/race specific number of jobs on change in log population in same commuting zone.

	N	No serial correlation correction		Control for serial correlation	
		$(lnJ_g)_{e,r,t}$	$(lnJ_g)_{e,r,t-3}$	$(lnJ_g)_{e,r,t}$	$(lnJ_g)_{e,r,t-3}$
White, NH	26000	0.2293***	0.2291***	0.0103***	0.0095***
		[0.0032]	[0.0032]	[0.0013]	[0.0012]
High School	8600	0.2144***	0.2128***	0.0071***	0.0063***
		[0.0041]	[0.0040]	[0.0014]	[0.0013]
Some College	8600	0.2332***	0.2333***	0.0105***	0.0097***
		[0.0039]	[0.0039]	[0.0013]	[0.0013]
College and Above	8500	0.2404***	0.2413***	0.0133***	0.0125***
		[0.0032]	[0.0032]	[0.0012]	[0.0011]
Black, NH	13000	0.0780***	0.0784***	0.0037***	0.0034***
		[0.0038]	[0.0038]	[0.0005]	[0.0005]
High School	4700	0.0783***	0.0781***	0.0019***	0.0016***
		[0.0046]	[0.0045]	[0.0006]	[0.0006]
Some College	4500	0.0925***	0.0932***	0.0049***	0.0045***
		[0.0050]	[0.0050]	[0.0006]	[0.0006]
College and Above	3800	0.0628***	0.0633***	0.0044***	0.0041***
		[0.0030]	[0.0030]	[0.0004]	[0.0004]
Hispanic	15000	0.0645***	0.0651***	0.0059***	0.0057***
		[0.0048]	[0.0049]	[0.0007]	[0.0006]
High School	6100	0.0705***	0.0706***	0.0049***	0.0047***
		[0.0055]	[0.0056]	[0.0006]	[0.0006]
Some College	5100	0.0733***	0.0740***	0.0066***	0.0064***
		[0.0052]	[0.0053]	[0.0007]	[0.0007]
College and Above	3800	0.0490***	0.0498***	0.0062***	0.0060***
		[0.0043]	[0.0043]	[0.0006]	[0.0006]

Notes: All regressions use the balanced panel, are weighted using CZ population shares, and include two-way (year and CZ) fixed effects. Robust standard errors are clustered at the CZ level. Correction for serial correlation indicates the inclusion of one-, two-, and three-year lags of the dependent variable as additional regressions. *, **, *** => statistical significance at the 90, 95, and 99 percent level, respectively. Sample includes 16-64 year-olds with at least a high school degree and observations from 2008-2019 due to the use of lagged variables.

Table 5 Marginal effects of a change in the log education/race specific number of jobs on change in log population in same commuting zone, at different points in the distribution of own-race population share.

		Own race population share in CZ, percentile		
	Overall m.e. of $(lnN_g)_{e,r,t}$ wrt change in $(lnJ_g)_{e,r,t-3}$	Marginal effect at the 25th Percentile of *OwnRaceShare*	Marginal effect at the 50th Percentile of *OwnRaceShare*	Marginal effect at the 75th Percentile of *OwnRaceShare*
M.E. for Whites, NH	0.0095***	0.0099***	0.0122***	0.0134***
	[0.0012]	[0.0013]	[0.0016]	[0.0017]
High School	0.0063***	0.0066***	0.0081***	0.0089***
	[0.0013]	[0.0013]	[0.0017]	[0.0018]
Some College	0.0097***	0.0101***	0.0125***	0.0137***
	[0.0013]	[0.0013]	[0.0017]	[0.0018]
College or Above	0.0125***	0.0131***	0.0161***	0.0177***
	[0.0011]	[0.0012]	[0.0014]	[0.0016]
M.E. for Blacks, NH	0.0034***	0.0008***	0.0018***	0.0050***
	[0.0005]	[0.0001]	[0.0003]	[0.0008]
High School	0.0016***	0.0004***	0.0008***	0.0024***
	[0.0006]	[0.0002]	[0.0003]	[0.0009]
Some College	0.0045***	0.0011***	0.0024***	0.0068***
	[0.0006]	[0.0001]	[0.0003]	[0.0009]
College or Above	0.0041***	0.0010***	0.0021***	0.0060***
	[0.0004]	[0.0001]	[0.0002]	[0.0006]
M.E. for Hispanics	0.0057***	0.0010***	0.0020***	0.0046***
	[0.0006]	[0.0001]	[0.0002]	[0.0005]
High School	0.0047***	0.0009***	0.0017***	0.0039***
	[0.0006]	[0.0001]	[0.0002]	[0.0005]
Some College	0.0064***	0.0012***	0.0023***	0.0052***
	[0.0007]	[0.0001]	[0.0002]	[0.0006]
College or Above	0.0060***	0.0011***	0.0021***	0.0048***
	[0.0006]	[0.0001]	[0.0002]	[0.0005]

Notes: See notes to Table 4. Number of observations are the same as in Table 4.

Table 6 Results from validity test of the effect of own race share on responsiveness.

	ME of change in race/ed-specific job changes on race/ed-specific population $\partial lnN_g\big\vert_{e.r.t} / \partial lnJ_g\big\vert_{e.r.t}$						
	Baseline (balanced) sample [1]	Validity (unbalanced) sample [2]	ME on regressor OwnRaceShare [3]	White Race Share [4]	Black Race Share [5]	Hispanic Race Share [6]	N
White, NH	0.0095***	0.0105***	0.1913***	**0.8940***	0.7995***	0.8001***	16000
	[0.0012]	[0.0013]	[0.0228]	[0.0955]	[0.1520]	[0.1405]	
Black, NH	0.0034***	0.0037***	0.2908***	0.5169***	**1.171***	0.6035***	14000
	[0.0005]	[0.0005]	[0.0419]	[0.0739]	[0.1456]	[0.1312]	
Hispanic	0.0057***	0.0059***	0.3577***	0.5821***	0.6515***	**0.9694***	14000
	[0.0006]	[0.0007]	[0.0398]	[0.0734]	[0.1533]	[0.1194]	
Specification:	Eq. (1)	Eq. (1)	Eq. (1)	Eq. (2)	Eq. (2)	Eq. (2)	

Notes: All estimations include the year-3 lag of log jobs, three years of lags of log population, are weighted by CZ population share and include both year and CZ fixed effects. Estimations in columns (1) and (3) are performed on the balanced sample that produce results for Table 4 and 5. Samples for results in columns (2) and (4)-(6) are unbalanced and eliminate CZs from the balanced panel that do not have representation of all racial groups.

Appendix A: Sample, Variable Construction, and Robustness Details

A1 Construction of Distance to Next Nearest Commuting Zone

The GEOcorr1990 data from the Missouri Census Data Center (https://mcdc.missouri.edu/applications/geocorr1990.html) provides the latitude and longitude for the centroid of each county in the U.S. We multiply each of these by the county population and divide by the county's CZ population, then sum over counties within the CZ. This renders population-weighted centroid for each commuting zone. These can then be used to calculate the distance between each commuting zones. As a regressor that is fixed in time, it is dropped from regressions that include CZ fixed effects.

A2 Construction of Migrant Shift-share Regressor

The regressions estimated include a regressor measuring the share of the population that are foreign-born (non-native). Because of the well-known existence of immigrant enclaves, this regressor could be endogenous to population growth and is, hence, instrumented using a Bartik-type instrument. The regressor included in the regression is constructed as follows (see Amior and Manning 2018a, Section D.3):

$$\mathrm{m}_{gt} = \frac{\sum_o \sigma^o_{g,t-1} N^F_{o(-g)t}}{N_{g,t-1}} \ , \qquad \text{(A1)}$$

where $\sigma^o_{g,t-1}$ is the share of the population in area *g* at time *t-1* that is native to origin country *o*; $N^F_{o(-g)t}$ is the stock of new origin-specific foreign migrants (excluding those living in area *g*) who arrived in the U.S. between *t-1* and *t*; this product is scaled by the initial total population of area *g*.

Shares of foreign-born population by county are obtained from IPUMS, NHGIS.[14] Five-year averages from 2005-2015 from the American Community Survey of the following tables are used:

Nativity in the United States: provides total number of native and foreign born in each county

Place of birth for the foreign-born in the United States: number of people by place of birth

The year of data in the analysis corresponds to the first year in the five years over which the population numbers are averaged. Data trends are extended backward one year (to allow for lagging) and forward through 2019 to match years of analysis. Due to small sample sizes, foreign-born are aggregated to the following geographic regions: Africa, Latin America, North America, Asia, Europe, and Oceania.

Since this regressor varies over both time and commuting zone, it remains in regressions including separate CZ and year fixed effects, but is dropped from robustness regressions including CZ/year combined fixed effect.

A3 Weighting Considerations

The use of weights is typically helpful for comparing estimates of different regressions from different samples and when the effect of interest varies with the size of the geographic units in the sample or there are heterogeneous effects. Weights may be called for here if the average population response varies with CZ size or with racial population size. In other words, is the same job opportunities more attractive in larger CZ than in smaller CZ, or in CZ with a different racial mix?

[14] Steven Manson, Jonathan Schroeder, David Van Riper, Tracy Kugler, and Steven Ruggles. IPUMS National Historical Geographic Information System: Version 16.0 [dataset]. Minneapolis, MN: IPUMS. 2021. http://doi.org/10.18128/D050.V16.0

When we apply differential weights by CZ or racial group size, the weighted regression puts greater weight on the largest CZ or the largest racial group. However, if the effect does not vary with CZ or racial group size, weighting becomes a less efficient regression strategy than using unweighted regression. When this happens, unweighted regression estimates may be more precise due to effective unweighted variation than weighted variation in the CZ sample (see T. J. Bartik and Sotherland 2019).

Additionally, the use of weights affects what the estimates of a regression represent. Weighted estimates show the average impact of a change in jobs on change in population for the United States, or, the average person in the U.S. When the change in jobs is modified by race/education, the estimates reflect the average impact for the average person of that race/education group in the U.S. Unweighted estimates show the average response for the average geographic unit in the sample.

Solon, Haider, and Wooldridge (2015) identify three reasons for weighting regressions: (1) to achieve more precise estimates in the presence of heteroskedasticity, (2) to achieve consistent estimates by correcting for endogenous sampling, and/or (3) to identify average partial effects in the presence of heterogeneous effects. Unweighted regression results are compared to those using weights, with no appreciable difference in estimated marginal effects.

In the unweighted regression, the null hypothesis of homoskedastic standard errors is rejected based on a Breuschh-Pagan test (results available upon request). While this may be reason enough to apply weights, since we are interested in heterogenous affects across CZs of widely varying sizes, Solon, Haider, and Wooldridge (2015) suggest that using weights may considerably improve the precision of the estimates.

Regarding endogenous sampling, Solon, Haider, and Wooldridge (2015) suggest if the

unweighted and weighted results are "similar," this would ease concern over endogenous sampling and, hence, weighting is unnecessary and may lead to less efficient estimates. The similarity in pattern of results in Table A1 between the unweighted and weighted marginal effects, suggests weighting is unnecessary. However, the presence of heteroskedasticity and interest in heterogenous effects leads us to report results from the weighted regression. We use weights constructed from CZ population shares for consistency with Amior and Manning (2018b).

Table A1: Marginal effects of a change in the log education/race specific number of jobs on change in log population in same commuting zone, comparing weighted (from Table 4) and unweighted regression results.

	N	Weighted Marginal Effect $lnJ_g\|_{e,r,t-3}$	Unweighted Marginal Effect $lnJ_g\|_{e,r,t-3}$
White, NH	26000	0.0095***	0.0401***
		[0.0012]	[0.0019]
High School	8600	0.0063***	0.0370***
		[0.0013]	[0.0020]
Some College	8600	0.0097***	0.0430***
		[0.0013]	[0.0021]
College and Above	8500	0.0125***	0.0403***
		[0.0011]	[0.0018]
Black, NH	13000	0.0034***	0.0122***
		[0.0005]	[0.0007]
High School	4700	0.0016***	0.0114***
		[0.0006]	[0.0007]
Some College	4500	0.0045***	0.0154***
		[0.0006]	[0.0008]
College and Above	3800	0.0041***	0.0096***
		[0.0004]	[0.0006]
Hispanic	15000	0.0057***	0.0106***
		[0.0006]	[0.0006]
High School	6100	0.0047***	0.0102***
		[0.0006]	[0.0007]
Some College	5100	0.0064***	0.0124***
		[0.0007]	[0.0008]
College and Above	3800	0.0060***	0.0087***
		[0.0006]	[0.0004]

Notes: Marginal effects are from the preferred specification (see notes to Table 4 in the text); weighted regression results are repeated from the last column in Table 4 in the text.

A4 Balanced vs. Unbalanced Panel

Balanced panels, especially in the presence of fixed-effects are generally preferred. When fixed effects are included, estimates from an unbalanced sample may be biased due to the incidental parameter problem (arising from a small number of time periods for certain race/education groups); for example, see Wooldridge (2010, chap. 10); also see FasterCapital (1995). Additionally, not knowing why some observations are missing in the unbalanced panel could expose the results to concerns about selection. And, making sure we have observations in each time period allows for the important lags included to account for serial correlation. Of course, restricting the sample to balanced observations may be throwing out relevant information. However, this is not a concern here since the results using both balanced and unbalanced samples are very similar.

The patterns of results seen in Table A2 are consistent with those presented in Table 5 using the balanced panel. Responsiveness generally increases in education and is weaker overall and within education levels among Black and Hispanic workers relative to White workers. This responsiveness, however, increases in own-race population shares. The marginal effect point estimates using the unweighted sample are slightly larger than those generated by the balanced panel, but not materially so.

Table A2 Marginal effects of a change in the log education/race specific number of jobs on change in log population in same commuting zone, at different points in the distribution of own-race population share, using unbalanced sample.

			Own race pop share in CZ, percentile		
	N	Overall m.e.	25th Percentile	50th Percentile	75th Percentile
M.E. for Whites, NH	26000	0.0113***	0.0118***	0.0146***	0.0160***
		[0.0013]	[0.0013]	[0.0017]	[0.0018]
High School	8600	0.0080***	0.0083***	0.0103***	0.0113***
		[0.0014]	[0.0014]	[0.0017]	[0.0019]
Some College	8600	0.0116***	0.0120***	0.0149***	0.0163***
		[0.0014]	[0.0014]	[0.0017]	[0.0019]
College or Above	8500	0.0145***	0.0151***	0.0186***	0.0204***
		[0.0012]	[0.0012]	[0.0015]	[0.0017]
M.E. for Blacks, NH	15000	0.0040***	0.0010***	0.0022***	0.0061***
		[0.0005]	[0.0001]	[0.0003]	[0.0008]
High School	5400	0.0022***	0.0006***	0.0012***	0.0034***
		[0.0006]	[0.0002]	[0.0003]	[0.0010]
Some College	5200	0.0052***	0.0013***	0.0028***	0.0080***
		[0.0006]	[0.0002]	[0.0003]	[0.0010]
College or Above	4300	0.0046***	0.0011***	0.0024***	0.0069***
		[0.0004]	[0.0001]	[0.0002]	[0.0006]
M.E. for Hispanics	18000	0.0061***	0.0011***	0.0022***	0.0051***
		[0.0007]	[0.0001]	[0.0002]	[0.0006]
High School	6900	0.0052***	0.0010***	0.0019***	0.0044***
		[0.0006]	[0.0001]	[0.0002]	[0.0005]
Some College	6400	0.0068***	0.0013***	0.0025***	0.0057***
		[0.0007]	[0.0001]	[0.0003]	[0.0006]
College or Above	4800	0.0063***	0.0012***	0.0022***	0.0051***
		[0.0006]	[0.0001]	[0.0002]	[0.0005]

Notes: Marginal effects are from the preferred specification (see notes to Table 4 in the text). These results are comparable to those presented in Table 5 in the text.